π-Board: Kanban-Style Alternative to Traditional To-Do Lists

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Abstract

Many to-do lists are either too complex or over-simplistic for personal use. Issue boards in the software domain address this challenge by integrating visualization with structured functionality to increase efficiency. Pi-board applies this Kanban-style approach for personal use by providing an open-source alternative.

Introduction

Kanban boards aim to visualize workflows and simplify the identification of bottlenecks by placing tasks into different columns according to their status. The benefits of Kanban has the following key aspects on workflow:

- Limits work-in-progress (WIP)
- Reduces task completion time
- Improves task prioritization

With π-board, benefits of Kanban can be applied to personal task management to support personal growth and success.

Literature

- Kanban positively affects **task prioritization** and decision-making through visualization and increased transparency.[1]
- Kanban increases **control over tasks** by capping work-in-progress and reducing average lead time. [1]
- Kanban enhances the quality of task completion through **workflow optimization**. [2]

Methods

The open-source π-board is built using Java, Maven, and requires a database, preferably PostgreSQL or Supabase, both of which are open-source. π-board implements a Kanban-style workflow visualization of personal tasks through the web-based user interface.

Beyond its standard functionality, π-board is designed to integrate seamlessly with local AI systems, such as Claude Desktop, via the MCP interface, enabling **AI supported** workflow assistance.

Visual Overview

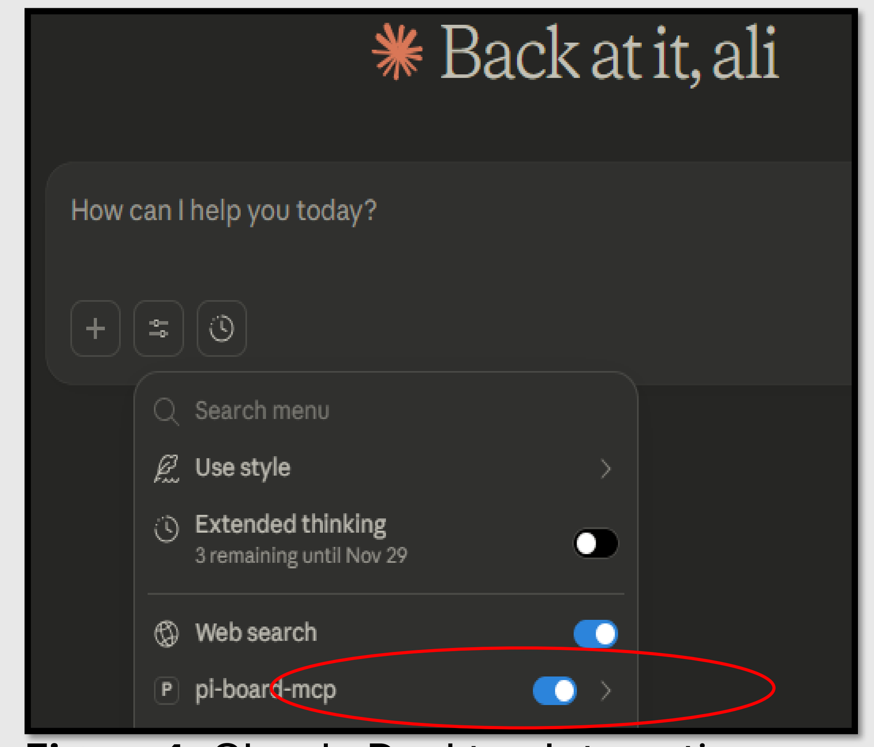

Figure 1: Claude Desktop Integration

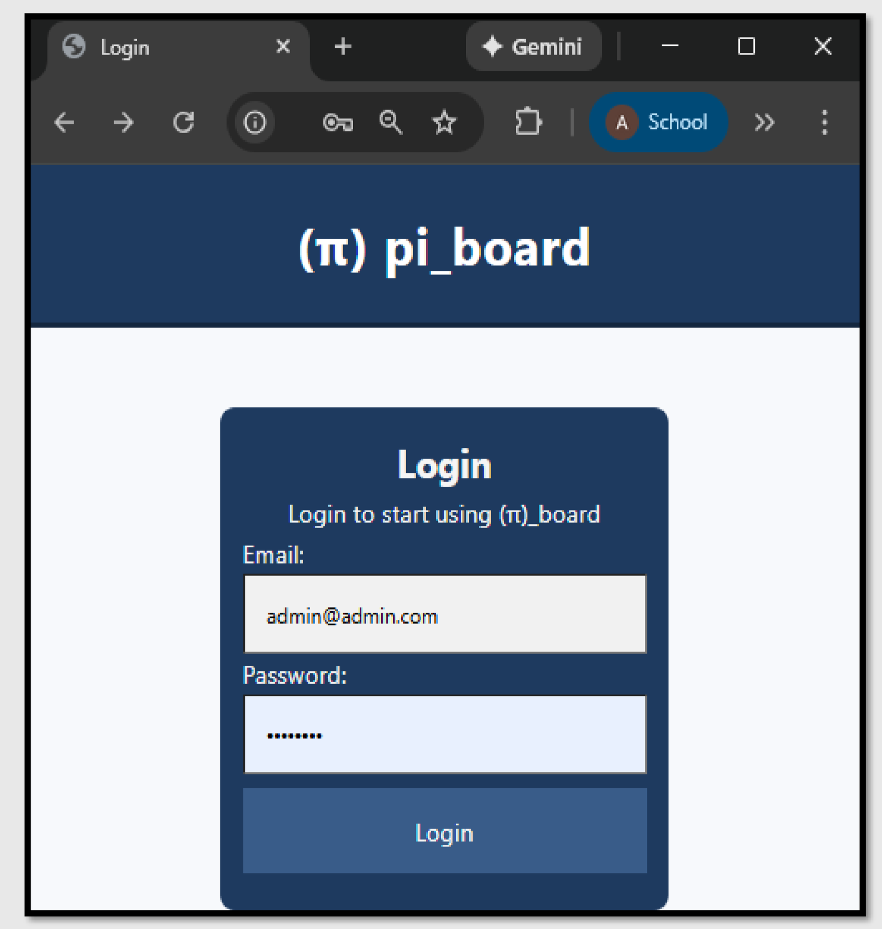

Figure 2: Web Interface

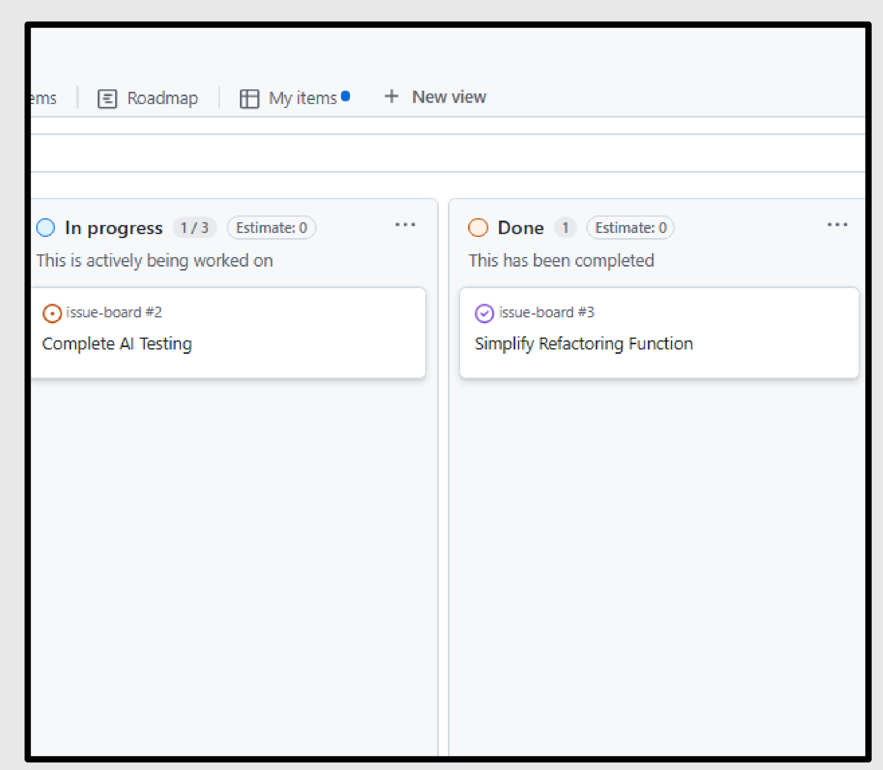

Figure 3: Kanban board from GitHub

Results

The implementation of π-board demonstrates that Kanban principles can be adapted for personal task management. By providing a visual interface of all tasks, users gain oversight of their responsibilities. By enabling AI integration, personal productivity can be further enhanced through a Kanban style alternative to traditional to-do lists.

Discussion

The project shows that Kanban-based visualization can be integrated with traditional to-do lists to improve personal productivity and task completion; however, its effectiveness ultimately depends on consistent user engagement.

References

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2) dos Santos, P. S., Beltrão, A. C., de Souza, B. P., & Travassos, G. H. (2018). On the benefits and challenges of using Kanban in Software Engineering: A structured synthesis study. Journal of Software Engineering Research and Development, 6(1). https://doi.org/10.1186/s40411-018-0057-1
3) Huang, C.-C., & Kusiak, A. (1996). Overview of kanban systems. International Journal of Computer Integrated Manufacturing, 9(3), 169–189. https://doi.org/10.1080/095119296131643